IV. Match the following.

Density	-	hρg
1 gwt	-	Milk
Pascal's law	-	$\frac{\text{Mass}}{\text{Volume}}$
Pressure exerted by a fluid	-	Pressure
Lactometer	-	980 dyne

V. Answer in brief.

1. On what factors the pressure exerted by the liquid depends on?
2. Why does a helium balloon float in air?
3. Why it is easy to swim in river water than in sea water?
4. What is meant by atmospheric pressure?
5. State Pascal's law.

VI. Answer in detail.

1. With an appropriate illustration prove that the force acting on a smaller area exerts a greater pressure.
2. Describe the construction and working of mercury barometer.
3. How does an object's density determine whether the object will sink or float in water?
4. Explain the construction and working of a hydrometer with diagram.
5. State the laws of flotation.

VII. Assertion and reason type questions.

Mark the correct answer as:

(a) If both assertion and reason are true and reason is the correct explanation of assertion.

(b) If both assertion and reason are true but reason is not the correct explanation of assertion.

(c) If assertion is true but reason is false.

(d) If assertion is false but reason is true.

1. **Assertion:** To float, body must displace liquid whose weight is equal to the actual weight.

   Reason: The body will experience no net downward force in that case.

2. **Assertion:** Pascal's law is the working principle of a hydraulic lift.

   Reason: Pressure is thrust per unit area.

VIII. Numerical Problems.

1. A block of wood of weight 200 g floats on the surface of water. If the volume of block is 300 cm^3, calculate the upthrust due to water.
2. Density of mercury is 13600 kg m^{-3}. Calculate the relative density.
3. The density of water is 1 g cm^{-3}. What is its density in S.I. units?
4. Calculate the apparent weight of wood floating on water if it weighs 100g in air.

IX. Higher Order Thinking Skills.

1. How high does the mercury barometer stand on a day when atmospheric pressure is 98.6 kPa?
2. How does a fish manage to rise up and move down in water?
3. If you put one ice cube in a glass of water and another in a glass of alcohol, what would you observe? Explain your observations.
4. Why does a boat with a hole in the bottom would eventually sink?



REFERENCE BOOKS

1. Fundamentals of Physics - By David Halliday and Robert Resnick.
2. I.C.S.E Concise Physics - By Selina publisher.
3. Physics - By Tower, Smith Tuston & Cope.



INTERNET RESOURCES

https://www.teachengineering.org/lessons/view/cub_airplanes_lesson04

http://www.cyberphysics.co.uk/topics/earth/atmosphr/atmospheric_pressure.htm

http://discovermagazine.com/2003/mar/featscienceof

http://northwestfloatcenter.com/how-flotation-can-help-your-heart/

IX_SCI_EM_Unit-03_PHY.indd 37 06-02-2020 11:38:20